1 Fig. 1.1 is a speed–time graph for an ice skater.

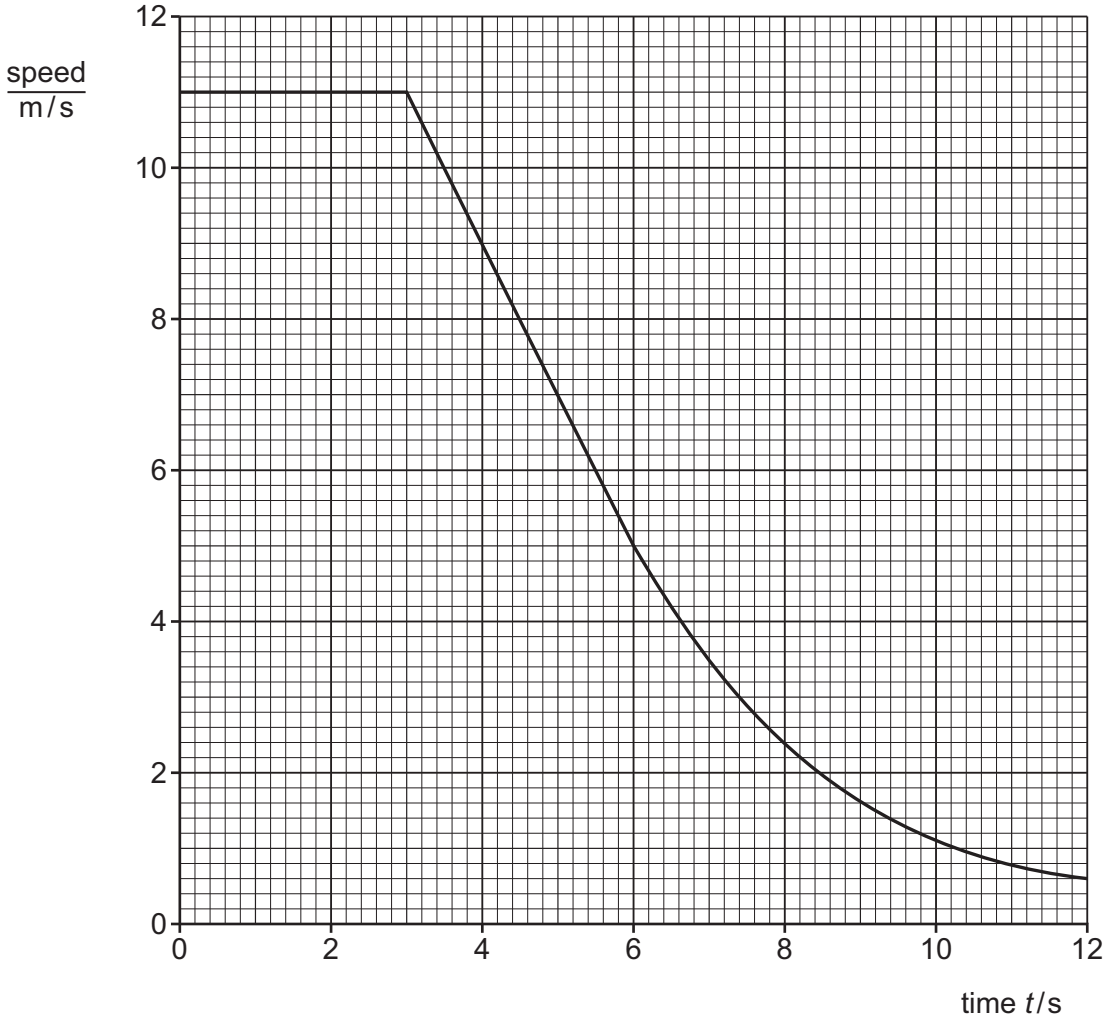

Fig. 1.1

(a) Describe the motion of the skater between $t = 0$ and $t = 3.0$ s.

..... [1]

(b) Calculate the distance travelled by the skater between $t = 0$ and $t = 3.0$ s.

distance = ... [2]

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(c) **(i)** State what is meant by deceleration.

..... [1]

(ii) Draw a tangent to the graph at $t = 9.0$ s.

Use the tangent to calculate the deceleration of the skater at $t = 9.0$ s.

deceleration = ... [3]

[Total: 7]

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